BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, PESHAWAR

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2021

CHEMISTRY

Past Paper 2021 • Paper Code: PR IX (001) 21

Retyped faithfully from the original board paper — same questions, no changes.

SECTION-A

Time Allowed: 15 Minutes | **Maximum Marks:** 12 | **Paper Code:** Chemistry Paper–IX (01) (21)

ا۔ ہر سوال کے سامنے چار دائرے دیے گئے ہیں۔ صرف صحیح جواب والا دائرہ بھر دیں۔ ۲۔ دائروں کو شیڈ (بھرنے) کے لیے نیلے یا کالے رنگ کا پین استعمال کریں۔ ۳۔ جواب میں ایک سے زائد دائرے بھرنے سے جواب غلط تصور ہوگا۔

Q1. One atmosphere is equal to.....

(A) 760 mm of Hg
(B) 790 mm of Hg
(C) 860 mm of Hg
(D) 880 mm of Hg

Q2. Which one is more concentrated solution:

(A) 0.5 Molar
(B) 0.05 Molar
(C) 0.005 Molar
(D) 1.0 Molar

Q3. The removal of Hydrogen is called.....

(A) Oxidation
(B) Reduction
(C) Oxidizing agent
(D) Reducing agent

Q4. .....is present in solid form.

(A) Chlorine
(B) Fluorine
(C) Iodine
(D) Bromine

Q5. Energy of 4S is smaller than.....

(A) 2S
(B) 2P
(C) 3P
(D) 3d

Q6. Oxidation number of S in H_2SO_4 is.....

(A) +4 (B) +5

(C) +6 (D) +7

Q7. Which type of bond exist in $CaCl_2$?

(A) Covalent (B) Ionic

(C) Metallic (D) Dative

Q8. The tendency of a metal to lose electron is called.....

(A) Electroplating (B) Electrolysis

(C) Electronegativity (D) Electropositivity

Q9. The symbol of copper is.....

(A) C (B) K

(C) Ca (D) Cu

Q10. P-Sub shell consists of.....orbitals.

(A) Two (B) Three

(C) Five (D) Seven

Q11. The modern periodic table is based on:

(A) Atomic number (B) Mass number

(C) Neutron number (D) Isotope number

Q12. An atom with a charge is called.....

(A) An electron (B) An ion

(C) A molecule (D) A metal

Section-B

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 32

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

QII. Answer any EIGHT parts. Each part carries FOUR marks.

1. Describe the contributions of Jabber Ibn Haiyan in the field of Chemistry.
2. Write electronic configuration for $^{29}_{14}Si$ and $^{40}_{18}Ar$.
3. Define atomic radius and explain its trends in periodic table.
4. Why do atoms form chemical bond?
5. Differentiate between diffusion and effusion.

6. Why NaOH is a strong but NH_4OH is a weak electrolyte?

7. Calculate the molarity of 50.0 cm^3 of solution containing 7.50 grams of $CaCl_2$.

8. Write down the chemical reaction of chlorine with $Ca(OH)_2$ and NH_3.

9. Differentiate between polar and non-polar covalent bond.

10. Write any four physical properties of Halogens.

11. What is Galvanic Cell?

Section-C

Time Allowed: 2 Hours 45 Minutes | **Maximum Marks:** 21

> Note: Note: Time allowed for Section – B and Section – C is 2 Hours and 45 minutes.

Q. Attempt any THREE questions. All questions carry equal marks.

QIII.

(a) [4] State and explain with example empirical formula and molecular formula of a compound.

(b) [3] Write the chemical formula for: (i) Methane (ii) Ammonia (iii) Sulphuric Acid

QIV.

(a) [4] Explain the Bohr's atomic theory.

(b) [3] Describe the fundamental particles of an atom.

QV.

(a) [5] What is vapour pressure? What are the factors which affect vapour pressure?

(b) [2] Why evaporation causes cooling effect?

QVI.

(a) [5] What do you mean by percentage composition? Write its types.

(b) [2] Differentiate between saturated and unsaturated solutions.